

**Factory Reallocation & Shipping
Optimization Recommendation System
for Nassau Candy Distributor**

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Abstract

Nassau Candy Distributor currently assigns products to factories using fixed rules, which can increase shipping time and logistics costs. This project develops a recommendation system that predicts shipping performance and suggests better factory assignments. Machine learning models are used to estimate shipping lead time, and different factory allocation scenarios are compared. The system also includes a Streamlit dashboard that allows users to perform what-if analysis and view recommendations. The proposed solution helps improve shipping efficiency while maintaining profitability.

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1. Introduction

Nassau Candy Distributor manufactures different candy products in multiple factories. At present, products are assigned to factories using traditional methods, which may not always be efficient. This can lead to longer delivery times, higher transportation costs, and lower profits.

The aim of this project is to develop a recommendation system that predicts shipping performance and suggests better factory assignments. This enables the company to make better business decisions before implementing any changes.

2. Problem Statement

The company faces the following problems:

- Products are assigned using fixed rules.
- Shipping distance is sometimes unnecessarily long.
- Delivery takes more time for certain regions.
- Transportation costs reduce profit.
- There is no system to test different factory assignments before implementation.

3. Objectives

The objectives of this project are:

- Predict shipping lead time.
- Compare different factory assignments.
- Recommend the best factory for each product.
- Improve shipping efficiency.
- Reduce logistics cost.
- Create an interactive Streamlit dashboard.

4. Dataset Description

The dataset contains information such as:

- Order ID
- Order Date
- Ship Date
- Ship Mode
- Customer Location
- Region
- Product Name
- Factory
- Sales
- Cost
- Gross Profit
- Units Sold

The project also includes factory coordinates and product-factory mapping.

5. Methodology

The project is completed in five steps:

5.1 Data Preparation

The raw dataset was cleaned and prepared before applying machine learning models. The following steps were performed:

- Removed missing and duplicate records.
- Converted date fields into useful features.
- Encoded categorical variables such as region and shipping mode.
- Normalized numerical features.

5.2 Business Rule-Based Lead Time Adjustment

During initial analysis, the calculated shipping lead time values contained unrealistic durations that did not represent actual business operations. To improve the reliability of predictions, business rules were applied to adjust abnormal lead time values.

The business rules considered factors such as:

- Shipping mode
- Distance between factory and customer region

- Typical delivery duration
- Operational feasibility

Extreme or unrealistic lead time values were corrected using predefined business constraints before training the machine learning models.

This step ensured that the predictive model learned realistic shipping patterns rather than incorrect historical anomalies.

5.3 Predictive Modeling

After preprocessing and business rule adjustments, machine learning models were trained to predict shipping lead time:

- Linear Regression
- Random Forest Regression
- Gradient Boosting Regression

5.4 Model Evaluation

The performance of the trained machine learning models was evaluated using standard regression evaluation metrics. The purpose of evaluation was to identify the model that provides the most accurate and reliable shipping lead time predictions.

The following metrics were used:

Mean Absolute Error (MAE):

Measures the average difference between actual and predicted lead time values.

Root Mean Square Error (RMSE):

Measures prediction error while giving higher importance to larger errors.

R² Score:

Represents how well the model explains the variation in shipping lead time.

The model with the best balance of prediction accuracy and interpretability was selected for further optimization and recommendation generation.

5.5 Route and Product Analysis

After selecting the best-performing model, shipping routes and product movements were analyzed to identify inefficient patterns.

The analysis focused on:

- Identifying routes with higher shipping delays.
- Finding regions with longer delivery times.
- Detecting product-factory combinations with poor performance.
- Understanding the relationship between factories, products, and customer regions.

Clustering techniques were applied to group similar shipping patterns and identify potential improvement opportunities.

5.6 Factory Reallocation Scenario Simulation

A scenario simulation engine was developed to evaluate alternative factory assignments for products.

For each product, different factory options were simulated and evaluated based on:

- Predicted shipping lead time.
- Expected operational improvement.
- Profit impact.
- Feasibility of reassignment.

The system compares the existing factory assignment with alternative factories and estimates the possible benefits before implementing any changes.

5.7 Recommendation Generation

Based on simulation results, the system generates factory reassignment recommendations.

Each recommendation is ranked using multiple factors:

- Reduction in shipping lead time.
- Profit stability.
- Operational risk.
- Improvement potential.

The final output provides decision-makers with the most suitable factory assignment options for improving supply chain performance.

5.8 Streamlit Dashboard Development

An interactive web dashboard was developed using Streamlit to present analytical results and recommendations.

The dashboard provides the following features:

- Product selection.
- Region and shipping mode filters.
- Factory optimization simulator.
- Current vs recommended factory comparison.
- Lead time improvement visualization.
- Profit impact analysis.
- Risk assessment panel.

The dashboard enables users to perform what-if analysis and make data-driven factory allocation decisions.

6. Results

The recommendation system provides:

- Faster shipping predictions
- Better factory allocation
- Reduced shipping delays
- Improved profit stability
- Easy comparison of different scenarios

8. Conclusion

This project improves the factory assignment process by using machine learning and optimization techniques. Instead of relying on fixed rules, the system recommends better factory assignments based on predicted shipping performance. The Streamlit dashboard makes it easy for users to analyze different scenarios and make informed decisions.

9. Future Scope

- Use real-time data for predictions.
- Improve accuracy using advanced AI models.
- Add demand forecasting.
- Optimize transportation costs further.